

PROJECT DESIGN PHASE

Team ID : NM2025TMID03763

Project Title : Calculating Family Expenses using ServiceNow Administrator

Project Design Phase: “Calculating Family Expenses Using ServiceNow”

1. Purpose

The purpose of this phase is to create detailed system designs, data models, workflows, and user interfaces for the family expense management solution in ServiceNow. This ensures a clear technical foundation before development begins.

2. Design Objectives

- Establish a scalable and efficient **data architecture** within ServiceNow.
- Design intuitive and user-friendly **forms, dashboards, and reports**.
- Create **automated workflows** for budgeting, expense approvals, and alerts.
- Define **role-based permissions** to ensure secure data access.
- Align all design elements with user needs and project objectives.

3. System Architecture

a. Components:

- **User Interface Layer:** Service Portal forms and dashboards for expense management.

- **Business Logic Layer:** Workflows, Flow Designer automation, and notification rules.
- **Data Layer:** Custom tables for expenses, budgets, and users.
- **Integration Layer (Future):** Optional API integrations with bank or finance apps.

b. Architectural Diagram (Conceptual)

[Users] → [Service Portal] → [ServiceNow Application Layer]

├— Expense Tables
├— Budget Tables
├— Workflows
└— Notifications

4. Data Model Design

Table Name	Purpose	Key Fields
Family Member [u_family_member]	Store family member info	Name, Role, Email
Expense Record [u_expense_record]	Capture individual expense entries	Date, Amount, Category, Description, Submitted By
Expense Category [u_expense_category]	Define categories for expenses	Category Name, Limit, Parent Category
Budget Threshold [u_budget_threshold]	Manage monthly/annual budgets	Month, Total Budget, Alert Level
Notifications [u_expense_alert]	Store alert and notification details	Type, Trigger Condition, Recipient

5. Workflow Design

Workflow 1: Expense Submission

1. User submits an expense via Service Portal form.
2. System validates data (amount, category).
3. Expense record is created in the Expense Table.
4. Notification sent to primary family account for visibility.

Workflow 2: Budget Monitoring

1. Scheduled job checks total monthly expenses.
2. If spending exceeds threshold → alert triggered.
3. Notification sent via email/SMS to family members.
4. Dashboard updates with over-budget indication.

Workflow 3: Monthly Report Generation

1. On the first of each month, system compiles previous month's data.
2. Generates summary by category and total spending.
3. Sends visual report link to family dashboard.

6. User Interface (UI) Design

a. Service Portal Pages

Page	Description
Home Dashboard	Overview of total expenses, budgets, and trends

Submit Expense	Form for logging new expenses
Expense History	List view with filters for date/category
Budget Settings	Configuration for monthly/annual budgets
Reports Page	Interactive charts and downloadable summaries

b. Visual Elements

- Cards for expense totals and category breakdowns
- Pie charts for expense distribution
- Line graphs for monthly trends
- Alert banners for overspending notifications

7. Security and Access Control

Role	Permissions
Admin (Parent)	Full access, can set budgets, view all expenses
Member (Child)	Can submit/view own expenses only
Viewer (Advisor)	Read-only access to reports and summaries

- Implement ACLs (Access Control Lists) on tables and forms.
- Use role-based notifications for alerts and workflows.

8. Design Validation Plan

- Conduct design walkthrough with stakeholders.
- Validate usability through wireframe feedback.
- Confirm that workflows meet budget tracking and approval logic.
- Ensure scalability for future integration (banking APIs or AI insights).

9. Tools & Technologies

- **ServiceNow Studio** (for tables, forms, and scripts)
- **Flow Designer** (for workflows and automations)
- **Performance Analytics** (for dashboards and reporting)
- **Service Portal Designer** (for UI customization)
- **Draw.io / Lucidchart** (for workflow and data model diagrams)

10. Expected Deliverables

Deliverable	Description
Data Model Diagram	Logical and physical design of tables
Workflow Diagrams	Visual representation of expense and budget flows
UI Mockups	Portal design prototypes
Security Matrix	Defined user roles and permissions
Design Review Document	Final approval of design artifacts

11. Next Steps

1. Conduct stakeholder review of design deliverables.
2. Finalize ServiceNow data tables and relationships.
3. Begin workflow automation scripting.
4. Prepare for **Development Phase** execution.